# Alienorum Calculation Reference

## Physical models, empirical estimates, and procedural appearance

Scientific audit edition · 13 November 2026

This reference describes the implemented mathematical procedures that turn catalog measurements and explicit scenario assumptions into object properties, climates, surfaces, atmospheres, and vegetation. Rendered equations are accompanied by units, parameter values, interpretation, and limitations.

Measured or supplied quantities take precedence where the corresponding calculation permits. A generated appearance is an illustrative realization. It is not evidence for an atmosphere, ocean, biosphere, mineral assemblage, or particular weather on an unresolved object.

The implementation itself is the authority for exact reproduction. The source locator and digest appendix ties this edition to the audited files. The formulas below describe scientific operations rather than software architecture.

Notation: SI units unless explicitly stated; M is mass in kilograms, R radius in metres, g acceleration in metres per second squared, p pressure in pascals, T temperature in kelvin, a semimajor axis, e eccentricity, and U an independent uniform variate on [0,1).

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# 1. Reading the models and preserving evidence

Implementation basis: planet.cpp; shore.cpp; numeric.h; savemodels.h.

Three levels of claim recur throughout this reference. Physical identities follow conservation, geometry, or equilibrium thermodynamics within stated approximations. Empirical estimates interpolate observations or published model grids. Scenario closures and procedural distributions fill gaps without establishing a unique physical solution.

Uncertainty annotations on a supplied number do not suppress the number. Explicit zero remains data; absence is different from zero. Catalog or editor pressure, composition, radius, temperature, and water coverage can constrain or bypass inference. Locked and saved appearances can retain earlier parameters, so a freshly inferred object and a restored object can legitimately differ.

$$C(x; l, h) = \min(h, \max(l, x)), \qquad S(x) = u^2(3 - 2u), \quad u = C(x; 0, 1)$$

$$\ell(x) = \frac{1}{1 + e^{-x}}, \qquad U(l, h) = l + (h - l)U$$

C denotes clipping; S is a smooth transition and  a logistic transition. Bounds in rendering are numerical or aesthetic limits unless explicitly tied to material physics. Scenario ranges are alternative assumptions, not statistical confidence intervals. A scientific reconstruction should retain input provenance, model diagnostics, object random state, and simulation epoch.

## Order of inference

Stellar properties and orbital illumination establish a forcing. Mass and radius establish gravity and escape speed. Formation inventories, release, escape, weathering, and condensation establish pressure and gas fractions. Radiation and phase partition are coupled. Terrain and retained water establish a waterline. Local climate and phase history establish exposed ice and liquid. Cloud and vegetation scenarios supply appearance; illumination and viewing geometry then alter the displayed colors.

The global irradiation path follows one resolved stellar light center, including the parent orbit for moons. It does not generally sum the radiative contribution of all stellar companions. Chemical abundances, age, delivery history, vertical mixing, interior storage, and aerosol properties remain substantial sources of model uncertainty.

# 2. Units, gravity, density, and basic astrometry

Implementation basis: celestial.cpp; point.cpp; star.cpp; planet.cpp.

$$\rho = \frac{3M}{4\pi R^3}, \quad g = \frac{GM}{R^2}, \quad v_{esc} = \sqrt{\frac{2GM}{R}}$$

The application stores many masses in grams; its internal gravitational constant compensates for that convention. Physical column calculations convert to kilograms. Confusing those conventions introduces a factor of one thousand. Surface gravity displayed by the common estimator is relative to Earth; atmospheric scale-height and inventory calculations use dimensional gravity.

$$R_{eq} = \frac{R}{(1-f)^{1/3}}, \qquad R_{pol} = (1-f)R_{eq}$$

Here f is oblateness. The conversion preserves the volume of an oblate spheroid. It is a shape convention, not a rotating interior calculation.

$$\mathbf{r} = d(\cos\delta\cos\alpha, \cos\delta\sin\alpha, \sin\delta)$$

$$d_{pc} = \frac{1}{\varpi_{arcsec}}, \qquad v_t = 4.74047\,\mu_{arcsec/yr} d_{pc}\ \mathrm{km\,s^{-1}}$$

Right ascension α and declination δ establish a direction. Coordinate rotations put it in the current view and reference plane. Proper motion and radial velocity provide a space velocity where available; generated populations use explicit velocity distributions described later. Linear propagation is a kinematic approximation, not an integrated Galactic orbit.

# 3. Stellar temperature, luminosity, and spectral approximations

Implementation basis: star.cpp; stellarevolution.h; wdcolor.h; wdlimb.h.

$$(B-V)(T) = 2.5\log_{10}\left[B_{\lambda_V}(T)/B_{\lambda_B}(T)\right] - c_{BV}$$

Temperature from B−V numerically inverts the Planck B/V ratio with the app's color zero-point correction c_BV. Bisection spans 1000–200000 K for 50 steps and clamps beyond the endpoint colors; supplied temperature and spectral tables can take precedence. Reddening, abundance, gravity, and unresolved companions can make color alone a biased thermometer.

$$M_{bol} = M_V + BC_V(T), \qquad \frac{L}{L_\odot} = 10^{0.4(4.74 - M_{bol})}$$

$$\frac{L}{L_\odot} = \left(\frac{R}{R_\odot}\right)^2 \left(\frac{T}{T_\odot}\right)^4$$

Bolometric correction uses the Flower/Torres polynomial in $\log_{10}(T)$ across its stellar range [R5]. Below 3500 K, a linear continuation joins −4.33 mag at 2500 K and −1.75 mag at 3500 K. Above 50000 K the V-band Planck-to-total-flux ratio continues the boundary value. These extensions are approximate, especially for molecular atmospheres and remnants.

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\left(e^{hc/(\lambda k_B T)} - 1\right)^{-1}$$

Main-sequence masses, radii, luminosities, temperatures, and colors are interpolated along the tabulated spectral sequence. They describe typical stars, not a unique inversion for an individual source. Evolution-grid inference compares normalized log-temperature, log-luminosity, and metallicity coordinates, using 64 nearby models and weighted summaries. Reported age ranges reflect that model procedure rather than a complete posterior over stellar history.

$$R_{WD} = 0.0126 R_\odot m^{-1/3}\sqrt{1-(m/1.44)^{4/3}}, \quad m = M/M_\odot$$

The white-dwarf radius approximation clips mass below 0.99 of 1.44 solar masses; missing mass takes 0.6 solar masses. Published atmosphere colors can supersede a blackbody approximation. Atmosphere type and cooling history affect the result.

# 4. Orbital geometry, rotation, and tidal constraints

Implementation basis: celestial.cpp; planet.cpp; moon.cpp; orbit_spacing.h.

$$n = \sqrt{\frac{G(M_c + M)}{a^3}}, \quad P = \frac{2\pi}{n}, \quad E - e\sin E = \mathcal{M}$$

$$r = a(1 - e\cos E), \quad (x, y) = a(\cos E - e, \sqrt{1 - e^2}\sin E)$$

The eccentric anomaly E is solved numerically. Rotations by the argument of periapsis, inclination, and ascending node place the ellipse in space. These are Keplerian trajectories; stability filters do not establish long-term many-body stability.

$$R_H = a(1 - e)\left(\frac{M}{3M_c}\right)^{1/3}, \quad R_{Roche} \simeq 2.44 R_c (\rho_c/\rho_s)^{1/3}$$

Satellite placement uses contact/Roche clearance and an outer fraction of the gravitational sphere of influence. Generated moons are limited to 0.3 of that scale. Ring particles use a fluid disruption scenario. Material strength, irregular shapes, resonances, and long-term tidal migration are not integrated.

$$P_{lock} = \frac{100}{1 + 4e}\ \text{days}, \qquad P_{spin} = \frac{2}{3}P_{orb} \quad (3:2)$$

Unknown rotation is assigned synchronous motion below the heuristic period threshold or for a hot giant. For periods below 250 days and e above 0.15, the next branch assigns a Mercury-like 3:2 spin resonance. This edition corrects its period ratio to 2/3. The threshold itself is a scenario rather than a tidal age calculation.

Other rotation estimates use A/ln(M expressed numerically in grams), with A equal to $2.06\times10^6$ for ocean-rich classes, $1.74\times10^6$ for ice giants, $1.11\times10^6$ for gas giants, and $2.38\times10^6$ otherwise, yielding seconds. The fixed gram reference is part of this empirical convention. It is not a dimensionally universal spin law.

$$\omega_{solar} = 2\pi\left(\frac{1}{P_{spin}} - \frac{1}{P_{orb}}\right)$$

# 5. Incident starlight, albedo, and equilibrium temperature

Implementation basis: planet.cpp; patch.cpp; star.cpp.

$$F_* = \frac{L/L_\odot}{(a/\mathrm{AU})^2}, \quad F(t) = \frac{F_*}{(1-e\cos E)^2}, \quad \langle F \rangle = \frac{F_*}{\sqrt{1-e^2}}$$

Fluxes F are relative to Earth; multiplication by 1361 gives W m⁻². The atmospheric retention calculation uses orbit-averaged flux. The baseline radiating temperature uses the stellar semimajor axis; seasonal forcing applies distance variation subsequently. For moons, the planet's stellar orbit supplies the baseline distance.

$$T_{eq} = \left[\frac{1361 F_* (1-A_B)}{4\sigma}\right]^{1/4}$$

The factor four is the ratio of spherical emitting area to intercepted disk area. Unit emissivity and global redistribution define this equilibrium temperature. It is not a measured surface temperature or the substellar temperature.

$$y = -\ln(1-A_g), \quad z = -\ln(1-A_B), \quad z = b_0 + b_1(B-V) + b_2 y$$

Bond albedo is estimated from a local plane through the three closest points in the transformed color/geometric-albedo data, then A_B=1−exp(−z), clipped to [0,1]. A degenerate triangle uses the nearest point. The eight calibration triples (B−V, A_g, A_B) are (0.97,.142,.088), (.81,.689,.760), (.20,.434,.294), (1.43,.170,.250), (.87,.538,.503), (1.09,.499,.342), (.56,.488,.300), and (.41,.442,.290). This sparse empirical interpolation is not a spectral integral and extrapolation can be fragile.

$$T_{rad}^4 = T_{eq}^4 + \frac{q_{tide}}{\sigma}$$

Internal tidal flux is added as energy, before taking the fourth root. The implemented path does not supply a general radiogenic or residual-cooling model for every object.

# 6. Tidal heat as an internal energy source

Implementation basis: tidalheat.h; planet.cpp.

The equilibrium-tide model uses zero spin obliquity and full eccentricity functions. Let z=e², b=1−z, and x=ω/n. The lag convention converts the assumed $k_2/Q$ at a representative forcing frequency. This is a present heating estimate; orbit and interior evolution are not coupled.

$$A = 1 + 3z + \frac{3}{8}z^2, \quad B = 1 + \frac{15}{2}z + \frac{45}{8}z^2 + \frac{5}{16}z^3$$

$$x_{eq} = \frac{B}{Ab^{3/2}}, \quad \Omega = \frac{A}{b^{9/2}}$$

$$H_{min} = \frac{z[3.5 + 11.25z + 28z^2 + (685/64)z^3 + (255/128)z^4 + (25/512)z^5]}{Ab^{15/2}}$$

$$q_{tide} = \frac{3(k_2/Q)GM_c^2R^3n}{4\pi a^6} \frac{H_{min} + \Omega(x - x_{eq})^2}{\max(1, 2|x-1|)}$$

An absent spin uses the minimum-dissipation equilibrium x_eq. The default $k_2/Q$ is 0.003 for solid-map worlds and $3\times10^{-6}$ for gaseous-map worlds. An explicit zero disables this heating. Rheology, ocean dissipation, obliquity tides, resonance forcing, and time-dependent Q remain unmodeled. At high eccentricity the model can predict large powers outside a plausible steady-state interior regime [R6].

# 7. Elemental abundances and formation composition

Implementation basis: chemistry.cpp; salinity.h.

$$[X/H] = \log_{10}(N_X/N_H) - \log_{10}(N_X/N_H)_\odot, \quad [X/H] = [X/Fe] + [Fe/H]$$

Abundance bundles retain their solar reference. Conversions account for those reference offsets. Missing elements can use a solar-scaled iron abundance in an explicitly marked formation scenario. This does not establish a planet's measured bulk composition.

$$n_X = 10^{\varepsilon_X - \varepsilon_{Si}}, \quad O_{rock} = \min(n_O, n_{Mg} + 2)$$

$$O_C = \min[\max(0, n_O - O_{rock}), 0.65 n_C], \quad O_W = \max(0, n_O - O_{rock} - O_C)$$

$$m_{rock} = 55.845 n_{Fe} + 24.305 n_{Mg} + 28.085 + 15.999 O_{rock}$$

$$m_W = 18.01528 O_W, \quad m_C = 12.011 n_C, \quad f_{ice} = \frac{m_W}{m_W + m_{rock}}$$

The oxygen allocation assumes a carbon partition of 45% CO, 10% $CO_2$, and 45% $CH_4$, which consumes 0.65 oxygen atoms per carbon. Silicate oxygen is assigned first. Water-to-rock and carbon-to-rock ratios, divided by the solar calculation, scale inventory scenarios. This stoichiometric closure omits detailed equilibrium condensation, oxidation states, carbon grains, disc transport, and migration [R7].

$$f_{Fe} = \frac{55.845 n_{Fe}}{55.845 n_{Fe} + 24.305 n_{Mg} + 28.085 + 15.999(n_{Mg} + 2)}$$

The iron-to-rock proxy is evaluated only in its supported abundance regime. It is a condensate proxy, not a directly observed core mass fraction. First-order error propagation uses variance $\Sigma(\partial f/\partial x_i)^2 s_i^2$ for independent abundance errors; correlations and systematic abundance offsets are outside that estimate.

# 8. Delivery inventories and atmospheric retention

Implementation basis: shore.cpp; shore.h.

The reference accessible inventories, as fractions of planetary mass, are $1.45\times10^{-4}$ $CO_2$-equivalent mass, $10^{-6}$ nitrogen, and $2.3\times10^{-4}$ water. Default age is 4.5 Gyr unless stellar age inference supplies another value. These are terrestrial scenario anchors, not universal primordial abundances.

$$a_s = 2.7\,C(M_*/M_\odot; 0.08, 10)\ \text{AU}, \quad w = \frac{1}{1+(a_s/a)^4}$$

$$D_W = \frac{0.001 + 0.02w}{0.001 + 0.02/[1+2.7^4]}, \quad D_M = C(M/M_\oplus; 10^{-4}, 100)^{1/4}$$

$$D_H = D_M \frac{1+\sqrt{D_W}}{2}, \qquad s_W = \frac{1}{1+2\sqrt{M/M_\oplus}}$$

Chemical water and carbon scales multiply the reference pools. Water delivery also receives D_M D_W and a factor three, with the surface fraction s_W normalized to one-third at Earth mass; carbon and nitrogen receive D_H. Maximum bulk fractions are 0.5 water, 0.05 $CO_2$, and 0.01 nitrogen. Present orbital position is only a weak proxy for formation location.

$$\mathcal{S} = 4\log_{10}\frac{v_{esc}}{5000} - \log_{10}\frac{\langle F\rangle}{0.43} + 0.6\log_{10}\frac{L}{L_\odot}$$

This diagnostic shoreline score is motivated by the empirical escape-speed-to-the-fourth relation [R1]. Positive values favor retention under the proxy. It does not uniquely specify surface pressure, and the inventory solver does not simply turn the score into a universal atmosphere probability.

# 9. Release, stellar activity, and escape history

Implementation basis: shore.cpp.

$$t_s = C[0.1(M_*/M_\odot)^{-2.5}A_Z; 0.01, 5]\ \mathrm{Gyr}$$

$$I(t) = t\ (t \le t_s), \qquad I(t) = t_s\left[1 + \frac{1-(t/t_s)^{-0.2}}{0.2}\right]\ (t > t_s)$$

Activity is saturated initially and then declines as $t^{-1.2}$. A_Z=10^(0.3[Fe/H]) applies only to main-sequence stars between 0.7 and 1.3 solar masses with a supported metallicity definition; otherwise it is one. This sensitivity is a scenario, not a universal magnetic-activity law.

$$D = \frac{I(t;t_s)}{I(4.5;0.1)}X, \quad \bar{A} = \frac{4.5D}{t}, \quad k = \min[10^{12}, 0.035\langle F\rangle\bar{A}(11200/v_{esc})^6]$$

$$\tau_r = C[0.15\sqrt{M/M_\oplus}; 0.01, 2]\ \mathrm{Gyr}$$

$$f_r = \frac{e^{-kt} - e^{-t/\tau_r}}{1 - k\tau_r}, \qquad f_r \to (t/\tau_r)e^{-t/\tau_r} \quad (k\tau_r \to 1)$$

Exponential mantle release into a reservoir with constant effective loss gives the retained fraction f_r. The stable numerical form avoids cancellation at equal rates. Initial inventory splits into unreleased, retained, and escaped mass. The heavy-gas loss coefficient is calibrated to broad Venus/Earth/Mars pressure scales; it is not an energy-limited hydrogen escape law.

$$k_W = k[1 + 500\ell((\langle F\rangle - 1.15)/0.025)]$$

The smooth steam transition raises water loss for strongly irradiated worlds. It is an effective historical scenario; cold-body water loss is set to zero in this closure. Age zero returns no released pool. Deep-water storage is separated before surface condensation.

# 10. Cold volatile worlds and nitrogen frost

Implementation basis: shore.cpp; shore.h.

The cold branch applies for bulk density below 3000 kg m⁻³ and orbit-averaged flux below 0.08 Earth flux. This is a regime selector rather than proof of an icy interior.

$$f_{ice} = C\left[\frac{1/\rho - 1/3500}{1/1300 - 1/3500}; 0.05, 0.7\right]$$

Specific-volume mixing uses effective densities of 3500 and 1300 kg m⁻³. Water inventory becomes f_ice; accessible nitrogen is min(0.01,1.35×10⁻⁴ f_ice/0.51 times the nitrogen abundance scale). Only 0.01 of the retained water is allocated to the surface. These choices are Titan-scale scenarios.

$$T_x = \max(2.7, 10^4\sqrt{\langle F\rangle \bar{A}}), \quad \lambda = \frac{0.028 v_{esc}^2}{2R_g T_x}$$

$$k_N = 1000\bar{A}\sqrt{T_x/1000}(1+\lambda)e^{-\lambda}\ \mathrm{Gyr}^{-1}$$

The exponent is bounded to avoid numerical overflow, and release time is 0.1 Gyr. This Jeans-shaped historical loss proxy does not calculate a measured exobase temperature or magnetospheric sputtering.

$$\log_{10}(p_{N_2}/10^5) = 3.7362 - \frac{264.651}{T - 6.788} \quad (T \geq 63.14\ \mathrm{K})$$

Below 63.14 K, continuous solid-phase fits use ln(p/bar)=8.514−458.4/T−19870/T²+480000/T³−4524000/T⁴ above 35.61 K, and 12.4−807.4/T−3926/T²+62970/T³−463300/T⁴+1325000/T⁵ below it. Multiplicative join factors match the branch boundaries. Below 10 K an exponential continuation tends monotonically toward vacuum. At 126.19 K and above the saturation cap is removed [R8].

$$T_\beta = 255\left[\frac{F_*(1-A_f)}{0.7\varepsilon_f}\right]^{1/4}, \quad T_\alpha = 3^{1/4}T_\beta$$

With A_f=0.8 and ε_f=0.67, the frost temperature uses the beta branch above 35.61 K, the alpha branch below it, and a 35.61 K coexistence plateau between them. An explicitly supplied cold temperature can override this estimate. Nitrogen collision absorption adds a bounded opacity based on p_N²/g; this is separate from the terrestrial gas-opacity terms.

# 11. Surface pressure and coupled gas–condensate partition

Implementation basis: shore.cpp; planet.cpp.

$$P_{col} = \frac{GM^2}{4\pi R^4}, \quad q_i = P_{col} f_i, \quad p = \sum_i q_i$$

$f_i$ is a planetary mass fraction and $q_i$ its hydrostatic column weight in pascals. The approximation holds g constant through a geometrically thin atmosphere. It is inappropriate as a detailed deep-envelope equation of state. The inferred pressure ceiling is $10^8$ Pa, a simulation bound.

$$p_{sat}(T) = p_0 \exp\left[\frac{L_m}{R_g}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right]$$

Water uses $(T_0, p_0) = (273.16\ \text{K}, 611.657\ \text{Pa})$, with molar latent heat 51060 J mol$^{-1}$ below that temperature and 45050 above; critical temperature is 647.10 K. $CO_2$ uses (216.59 K,518500 Pa), latent heats 25000/16000 J mol$^{-1}$ and critical temperature 304.13 K. These constant-latent-heat expressions are approximations, especially near critical conditions. Effective mixed-column water humidity multiplies the water saturation pressure by 0.15.

$$q_i^{gas} = \min\left(q_i^{available}, p_{sat,i}\frac{\mu_i}{\bar{\mu}}\right), \quad \bar{\mu} = \frac{\sum_i q_i^{gas}}{\sum_i q_i^{gas}/\mu_i}$$

Column weights are not mole partial pressures. Simultaneous solution for mean molar mass prevents sequential condensation from leaving inconsistent partial pressures. The tracked inferred pools are $CO_2$, $N_2$, and $H_2O$, with approximate molar masses .044, .028, and .018 kg mol$^{-1}$. Gas mole fractions are proportional to $q_i/\mu_i$.

$$w_T = \ell((T - 273.15)/5)[1 - \ell((T - 360)/10)]^{\frac{f_W}{f_W + 10^{-6}}}$$

$$f_{C,air} = \exp[-3t\, w_T \eta_W \min(1, \sqrt{M/M_\oplus})]$$

Wet weathering moves carbon from the accessible atmospheric pool into a protected reservoir. Candidate temperatures are tested against the radiative temperature; a scan over 128 intervals brackets sign changes, followed by bisection. Stable converged roots are preferred near a heuristic climate anchor. Multiple equilibria and residuals are retained in diagnostics. A supplied temperature is a constraint and can disagree with radiative balance; that disagreement is recorded rather than hidden.

# 12. Thermal opacity and the Titan calibration

Implementation basis: misc.cpp; shore.cpp; planet.cpp.

Let P be pressure in terrestrial atmospheres and $p_i = P\,x_i$ the corresponding mole partial-pressure proxy. The gray optical-depth closure sums logarithmic gas terms, a hydrogen feedback on methane, and collision terms. Its coefficients are empirical; it is not line-by-line radiative transfer.

$$\tau_i = a_i \ln(1 + b_i p_i), \qquad h_{H_2} = 1 + 25 p_{H_2}$$

| Species | $a_i$ | $b_i$ | Sign / qualification |
|---|---|---|---|
| $CO_2$ | .5 | 100 | positive |
| $CH_4$ | 1.2 | 500 $h(H_2)$ | positive |
| $H_2O$ | .8 | 50 | positive |
| $N_2O$ | .7 | 350 | positive |
| $O_3$ | .4 | 150 | positive |
| $SO_2$ | .5 | 80 | positive |
| $H_2S$ | .15 | 20 | positive |
| CO | .05 | 10 | positive |
| HCN | .1 | 30 | positive |
| $NH_3$ | 1.7 | 800 | positive |
| $C_2H_6$ | .20 | 600 | subtracted calibration term |

$$\tau = \max\left(0, \sum_{i \neq C_2H_6} \tau_i - \tau_{C_2H_6} + 0.00719 p_{CO_2}^2 + \tau_{H_2,\,CIA}\right)$$

$$\tau_{H_2,\,CIA} = 0.08 p_{H_2} P \quad (P > 0.5), \qquad T_s = T_{rad}(1 + 1.8\tau)^{1/4}$$

The subtracted ethane term is retained to preserve the app’s Titan temperature calibration. Ethane itself has infrared absorption [R4]; therefore this negative term must be read as an effective net cooling proxy in the combined empirical closure, not as negative molecular absorption. The code does not explicitly derive Titan’s photochemical aerosol production or solve its shortwave anti-greenhouse balance. A future separated gas/aerosol energy model would require recalibration against Titan. The term was not changed in this edition.

Hydrogen collision absorption switches on at P=0.5; this creates a small formula discontinuity when $H_2$ is present. $CO_2$ collision absorption is calibrated near Venus and scales quadratically. The coefficients fold in approximately terrestrial gravity; their transfer to other gravities is limited. At very large warming $\Delta = T_s - T_{rad}$, values above 500 K are mapped to $500 + 100 \tanh((\Delta - 500)/100)$. This is an extrapolation guard, not a runaway-greenhouse criterion.

# 13. Primordial hydrogen–helium envelopes and object classification

Implementation basis: shore.cpp; chemistry.cpp; planet.cpp; classification_model.json.

$$R_c = 1.02 R_\oplus m^{0.27}, \quad f_0 = C[0.015(m/5)^{1.5}; 10^{-5}, 0.2]$$

$$R_{env} = 2.06 R_\oplus m^{-0.21}(f/0.05)^{0.59} F^{0.044} (\max(0.1, t)/5)^{-0.18}$$

m=M/M⊕. The envelope thickness relation is based on Lopez and Fortney [R2]; its published model domain is 1–20 Earth masses, 0.01–20% envelope mass, 0.1–1000 Earth irradiation, and 0.1–10 Gyr. Application outside that domain is an explicitly approximate extension.

$$R_X = \max(2R_c, R_c + R_{env}(f_0, 0.1)), \quad \Delta f = \frac{0.1\pi R_X^3 \mathcal{E}_{XUV}}{GM^2}$$

$$\mathcal{E}_{XUV} = 1361 F\, 10^{-3} I(t) X \text{ (Gyr converted to seconds)}$$

$$\lambda_H = \frac{GM(0.0023)}{R_g T_{eq} R_X}, \quad f = \max[0, f_0 \ell((\lambda_H - 20)/3) - \Delta f]$$

The fixed expanded early radius, 0.1 heating efficiency, and saturated XUV fraction $10^{-3}$ are scenarios. Core-powered mass loss, tidal escape corrections, and a coupled cooling trajectory are absent. The retained score is an initial envelope score times f/(f+.001); the 0.5 threshold chooses a class only when evidence and type locks permit it. A measured radius can override that choice.

$$P_{base} = \frac{GM^2 f}{4\pi R_c^4}, \quad R = R_c + R_{env}$$

The empirical large-radius classifier uses logistic scores on $\log_{10}$ mass, $\log_{10}$ irradiation, and metallicity. Current abundance-aware coefficients are (−1.025614,10.724353,−2.900638,2.057835); baseline coefficients are (−1.652304,10.917212,−2.717543). Its training target contrasts radius at least 2 Earth radii with at most 1.4, rather than measuring gas composition. The domain is 2.55–29.32 Earth masses, 2.49–2290 Earth flux, 4900–6300 K hosts, and metallicity −.44 to .439 dex. Class-balanced scores are not occurrence probabilities. A separate cool-formation prior avoids an unrestricted extrapolation of the hot-sample fit.

# 14. Bombardment, erosion, and accessible volatile delivery

Implementation basis: shore.cpp; celestial.cpp.

$$B = C\left[\sqrt{30\mathrm{AU}/a_{belt}}; 1, 4\right], \quad f_{imp} = \min[0.01, 10^{-5}(B-1)X_{imp}t/4.5]$$

Absent a usable debris-belt inner edge, B=1. Thus this term represents excess over the reference bombardment history, not the whole impact history. Reference impactors contain mass fractions .1 water, .01 $CO_2$, and .001 nitrogen.

$$q^2 = \min\left(10^6, \frac{GM_*}{av_{esc}^2}\right), \quad c = \frac{1}{1+q^2}, \quad \varepsilon = \min(1, 0.03q^2)$$

$$d_g = \frac{0.9f_{imp}\varepsilon}{f_{gas}}, \quad d_s = \frac{0.1f_{imp}\varepsilon}{f_{mobile}}, \quad V(d) = \frac{1-e^{-d}}{d}$$

A midpoint reference partition allocates 90% of erosion capacity to atmospheric gas and 10% to exposed mobile condensates. Original pools lose fractions weighted by 1−exp(−d); protected carbon and deep water are excluded. Arriving material is captured with fraction c; delivery distributed across the history survives with V of the combined erosion/escape exposure. Its continuous limit is V(0)=1. Species diagnostics distinguish prompt loss, later loss, retained delivery, and original erosion. These are effective exposure assumptions, not hydrodynamic impact calculations.

Crater abundance separately decreases with atmospheric pressure, through 1/(1+3p/p⊕), and increases with B. That appearance suppression does not suppress physical bombardment in the volatile budget.

# 15. Water inventory, basin volume, and surface coverage

Implementation basis: shore.cpp; watercoverage.h; planet.cpp.

$$d_W = \frac{f_{W,\,surface} M}{4\pi R^2 \rho_W}, \quad \rho_W = 1000\ \mathrm{kg\,m^{-3}}$$

$$\sigma_h = \min(0.05R,\, 2500 \times 9.80665/g)$$

The default basin relief is 2500 m at Earth gravity. It scales inversely with gravity as a yield-limited scenario, with a 5% radius cap. Physical basin relief is intentionally separate from visual displacement. Otherwise shallow decorative terrain would flood with very little physical water. Retained condensed inventory includes ice; this coverage is not a detection of exposed liquid.

$$d_W = \frac{\sigma_h}{s_h}\frac{1}{N}\sum_j \max(0, h_0 - h_j), \quad f_{area} = \frac{1}{N}\sum_j \mathbf{1}_{h_j < h_0}$$

For equal-area normalized terrain samples h_j of standard deviation s_h, bisection finds the waterline $h_0$ by matching volume. A flat surface with positive water is fully covered. The stored visual waterline is clipped to [0,1], while the physical represented depth can exceed the highest sampled relief.

$$\frac{d_W}{\sigma_h} = z\Phi(z) + \phi(z), \quad f_{area} = \Phi(z)$$

Before terrain exists, Gaussian hypsometry supplies an initial coverage and the visual level is clipped from 0.5+0.08z. Sampled elevations later calibrate that level. A percentage editor measures equal-area coverage; the waterline itself is not generally equal to the covered fraction.

# 16. Salinity, freezing, boiling, and exposed liquid

Implementation basis: salinity.h; surfaceclimate.h; surfacephase.h.

$$r_s = \frac{35}{965}\frac{Z_s}{W}, \quad S_{ocean} = \min\left(264, \frac{1000 r_s}{1 + r_s}\right)\ \mathrm{g\,kg^{-1}}$$

$$Z_s = 10^{C[0.5([Na/H] + [Cl/H]) - [Fe/H];\ -4,\ 4]}$$

The salt inventory is an Earth-calibrated mobilization scenario. Na can use Mg as a proxy; Cl can use S or Ar. With no constraining abundance ratios, salt scaling is one. W is the relative liquid-water scale with a positive numerical floor. A supplied salinity takes precedence; a solar-scaled metallicity alone does not make oceans arbitrarily saltier. The result is not an ion-by-ion ocean analysis.

$$T_f = 273.15 - 0.0575S + 0.001710523S^{3/2} - 0.0002154996S^2 - 0.000753p_{dbar}$$

The seawater polynomial uses S clipped to 120 g kg⁻¹ and pressure above reference converted to decibars. Beyond that salinity the app linearly approaches a 252.05 K NaCl-brine scenario by 233 g kg⁻¹. This extension and very high pressure are outside a full phase diagram; high-pressure ice polymorphs and specific dissolved mixtures are absent.

$$T_b = \min\left[647.096, \left(\frac{1}{373.15} - \frac{8.314}{40660}\ln\frac{\max(1, p)}{10^5}\right)^{-1}\right]$$

When the inverse is nonpositive, the boiling estimate is 647.096 K. This constant-latent-heat saturation approximation does not distinguish a supercritical fluid as liquid water. Exposed ocean water is suppressed when pressure/temperature permit evaporation; an ice-covered reservoir can retain subsurface liquid. Land frost and open water use separate response rules.

# 17. Latitude, altitude, day–night contrast, and regional climate

Implementation basis: climate.h; planet.cpp.

$$P = p/101325, \quad D = T_m[0.04 + 0.25/(1+P)], \quad \eta = P/(P+0.1)$$

$$\Gamma = 0.0065C(g/g_\oplus; 0, 3)\frac{P}{P+0.01}, \quad T_{lat} = T_m + D(1/3 - z^2)$$

z is sine latitude, so its equal-area mean square is 1/3. The latitude term preserves the baseline mean before land warming and nonlinear seasonal modifications. η is the redistribution proxy and Γ is an approximate lapse rate in K m⁻¹. It is not a full moist adiabat or a measured lapse rate for each atmosphere.

$$T_{sync} = T_m[f_{int} + (1 - f_{int})(\eta + 4(1-\eta)\max(0, \mathbf{n}\cdot\mathbf{s}))]^{1/4}$$

$$\Delta T_{dry} = 0.04 T_m \frac{P}{P+0.01}\left|\max\left(0, 1 - \left(\frac{z^2 - 0.18}{0.2}\right)^2\right)\right|^2$$

The dry-belt warming acts on non-synchronous land. A spherical regional field combines sinusoids with weights .65, .25, and .10 and amplitudes T_m×.012/(1+P/10). For coordinates (x,y,z), its factors are sin(3x+1.7y)sin(4z−2y), sin(13x−7y+3z)sin(11z+5y), and sin(43x+29y)sin(37z−17y). These break up boundaries without seams; they are a climate scenario, not weather data. Contrast, dry warming, and regional amplitude all scale by 1−f_int. Elevation above sea level subtracts Γh and temperatures have a 2.7 K floor.

# 18. Seasonal and diurnal thermal response

Implementation basis: thermalclimate.h; planet.cpp; climate.h.

$$C_A = c_{p\overline{g}}^{p}, \quad C_L = 2 \times 10^6 + C_A + 0.3 f_o \eta (2.09 \times 10^8)$$

$$C_O = C_A + 2.09 \times 10^8, \quad \tau = C/B, \quad B = \max[0.01, (1361 F_* (1 - A_B) + 4 q_{tide})/T_m]$$

Heat capacities are J m$^{-2}$ K$^{-1}$, B is W m$^{-2}$ K$^{-1}$, and τ is seconds. The ocean term corresponds to roughly 50 m of water. f_o is equal-area water coverage. Mixture heat capacity uses molar fractions and gas molar heat capacities; missing mixture takes approximately terrestrial air. Transport and mixed-layer depths remain scenario choices.

$$Q_d(\varphi, \delta) = \frac{H \sin\varphi \sin\delta + \cos\varphi \cos\delta \sin H}{\pi}$$

$$H = \arccos C(-\tan\varphi \tan\delta;\ -1, 1)$$

Daily mean projected insolation handles polar day/night by clipping the sunset angle. Annual heating is sampled at 64 phases. Thermal history uses 128 samples weighted by exp(−lag/τ) times duration, normalized to unity. Short memory samples uniform time through up to ten response times; longer memory samples eccentric anomaly with its correct duration weight. This quadrature is periodic and independent of display frame rate.

$$A_n^{out} = \frac{A_n^{in}}{\sqrt{1 + (n\omega_{solar}\tau)^2}}, \quad \phi_n = \tan^{-1}(n\omega_{solar}\tau)$$

Eight harmonics of clipped daytime illumination supply diurnal forcing. Land uses (2×10$^5$+0.05C_A)/B; water blends 4.18×10$^6$ and 1.93×10$^6$ J m$^{-2}$ K$^{-1}$ according to a 263.15–283.15 K liquid transition, plus 0.05C_A. Stellar modulation changes the external part of T$^4$; internal heat remains constant. A supplied mean temperature uses a √(1−e$^2$) normalization so orbital eccentricity does not silently redefine that supplied mean.

# 19. Sea ice as an enthalpy reservoir

Implementation basis: iceclimate.h; surfaceclimate.h; planet.cpp.

$$L_A = \rho_W D L_f, \quad f = C(-E/L_A; 0, 1), \quad h = fD\rho_W/\rho_I$$

E is energy per unit area relative to the unfrozen freezing-point reservoir. L_f=334000 J kg⁻¹, ρ_W=1000 and ρ_I=917 kg m⁻³. Negative enthalpy represents latent heat in ice; fully frozen columns can cool sensibly. A default depth/mixed layer of 50 m is a scenario. Ice conductivity is 2.2 W m⁻¹ K⁻¹ [R3].

$$C_W = C_A + 4.18 \times 10^6 \min(D, D_{mix}), \quad C_I = C_A + 1.93 \times 10^6 \min(2, D\rho_W/\rho_I)$$

$$T = T_f + E/C_W \quad (E \geq 0), \qquad T = T_f(1) + (E + L_A)/C_I \quad (E \leq -L_A)$$

Between those regimes, brine rejection changes T_f as the liquid fraction shrinks. Salt-to-water ratio increases by the inverse remaining water fraction, capped at 264 g kg⁻¹. Surface temperature of partially frozen water follows a conductive balance constrained not to exceed the local freezing temperature.

$$c_{ice} = 1 - e^{-h/0.1}, \quad a_{clim} = \tanh(h/0.5)$$

Optical whiteness and climatic albedo use different thickness scales. Water/ice albedos are .07/.6 with default visibility .5. Albedo changes shortwave absorption only; transported heat remains energy input. Ocean heat transport is partitioned between the ice base and atmosphere while preserving the total.

$$E_{new} - E_{old} = \Delta t F_{net}(E_{new})$$

Implicit energy integration crosses freezing/melting boundaries without instantaneous phase switching. Periodic histories use 17 sine-latitude rows and 128 orbital phases, with longitude resolution on synchronous worlds. Convergence diagnostics report the periodic closure. Thin-frost fallback uses a 10 W m⁻² K⁻¹ exchange, 10 cm of ice for water or 1 cm for land, and a ±0.15 K dead band. That fallback is distinct from the finite ocean enthalpy model.

# 20. Habitability flags, vegetation habitat, and radiation belts

Implementation basis: planet.cpp; vegetation.h; habitablezone.h.

$$S_{eff} = S_0 + a\Delta T + b\Delta T^2 + c\Delta T^3 + d\Delta T^4, \quad \Delta T = C(T_*; 2600, 7200) - 5780$$

Conservative habitable-zone limits follow the Kopparapu runaway/maximum-greenhouse polynomial [R9]. Inner coefficients are interpolated linearly between 0.1, 1, and 5 Earth masses; outer coefficients are mass-independent. Endpoint approximation applies outside that mass range. This is a 1D water-based zone, not a life detection.

| Boundary / mass | $S_0$ | a, b, c, d |
|---|---|---|
| Inner / 0.1 | .99 | 1.209e−4, 1.404e−8, −7.418e−12, −1.713e−15 |
| Inner / 1 | 1.107 | 1.332e−4, 1.580e−8, −8.308e−12, −1.931e−15 |
| Inner / 5 | 1.188 | 1.433e−4, 1.707e−8, −8.968e−12, −2.084e−15 |
| Outer | .356 | 6.171e−5, 1.698e−9, −3.198e−12, −5.575e−16 |

The legacy global surface-habitability flag combines this orbital zone, mass above .02 Earth masses, waterline at least .05, pressure from 600 Pa to below 2000 atmospheres, and mean temperature between .9×273.15 K and 320 K. These are explicit classification gates. A waterline threshold is not a rigorous minimum area of exposed liquid, and the 245.8 K lower temperature gate does not by itself establish liquid water.

$$P_{belt}(d) = 0.01 + \frac{0.99}{1 + (d_0/d)^s}$$

For moons of gaseous worlds, d is distance in parent equatorial radii. The transition $d_0$=(8+12U) times mass and spin factors. Mass factor is clipped (M_parent/M_J)^(1/6) to [.6,1.8]; spin factor is clipped (10 h/|P_spin|)^(1/6) to [.7,1.4]. Slope s=3+2U. Thirty-two eccentric-anomaly samples average this survival score with duration weights 1−e cos E.

$$P_{shield} = 1 - (1 - \langle P_{belt} \rangle)\exp[-(p/g)/(101325/9.80665)]$$

A stable object-specific draw decides simulated surface life against this score. This is a procedural radiation-belt scenario, not a calculated magnetic field, charged-particle spectrum, dose in grays, or biological survival probability. It does not heat the climate. Atmospheric column mass supplies shielding heuristically. Other orbital parents take score one in this particular rule.

# 21. Vegetation adaptation and photosynthetic reflectance

Implementation basis: vegetation.h; color.cpp.

$$a_v = 0.55 + 0.30U, \quad T_c = 294 + a_v[(T_{cold} + T_{hot})/2 - 294] + U(-8, 8)$$

$$\Delta T_v = C\{[16 + a_v((T_{hot} - T_{cold})/2 - 16)](0.65 + 0.45U); 12, 60\}$$

A climate survey samples 13 sine-latitudes, 24 longitudes, sea level, and representative uplands. Adapted minimum/maximum temperatures are T_c±ΔT_v, clipped to 223.15–453.15 K. Those deliberately generous speculative biochemical limits are not measured universal growth limits. Habitat evaluation samples the year and day, with 33 latitude rows, 16 longitude columns for synchronous worlds, and eight diurnal phases otherwise. At least one tenth of samples must fall inside the adapted range, the liquid phase window, and the biochemical bounds.

$$w_\lambda = 0.20 + 0.80\frac{\lambda B_\lambda(T_*)}{\max(\lambda B_\lambda(T_*))}$$

Pigment centers are drawn on 350–1000 nm at 5 nm steps using these photon-weighted relative probabilities plus ±2.5 nm jitter. One center is constrained to the visible 400–700 nm range. Total band count is two through five. The first two widths are U(10,25) nm; accessory widths U(20,45) nm. Common concentration is log-uniform from .6 to 5 and each band depth multiplies U(.5,1.5).

$$\tau_p(\lambda) = \sum_j d_j \exp\left[-\frac{(\lambda - \lambda_j)^2}{2s_j^2}\right]$$

$$R(\lambda) = C\left[\left(r_V + \frac{r_{IR} - r_V}{1 + e^{(\lambda_e - \lambda)/w_e}}\right)\frac{e^{-\tau_p(\lambda)}}{1 + e^{(\lambda_{UV} - \lambda)/10}}; 0, 1\right]$$

Visible scattering r_v=U(.15,.65), infrared scattering min(.90,r_v+.25), UV edge U(320,380) nm, and structural edge max(700,max_j(λ_j+2s_j)). This is a hypothetical leaf-like reflectance spectrum. It does not calculate a real alien pigment or oxygenic photosynthetic efficiency. RGB is the average reflectance in overlapping windows 560–709, 480–599, and 400–509 nm. It preserves dark absorption rather than normalizing every spectrum to brightness one.

An artistic population rule favors purple over green by 5:2 for saturated candidates in those hue families. It can resample up to 128 pigment mixtures; other hue families pass through. That selection is not an empirical prediction of extraterrestrial ecology. Illumination tint is applied separately from intrinsic reflectance.

# 22. Trunks, crown dimensions, foliage texture, and placement

Implementation basis: plantform.h; plantmesh.h; planttexture.h; visuals.cpp; surfacescatter.h.

$$g_v = C(g/g_\oplus; 0.1, 10), \quad p_v = C\left[\frac{\ln(1+\max(0, p/p_\oplus))}{\ln 2}; 0.25, 4\right]$$

$$H_t = 10/g_v, \quad W_t = 0.3 g_v, \quad R_c = \min(2.5/p_v, 10), \quad o_c = 0.8 p_v$$

These multipliers act on each local scenery size. The crown vertical radius is horizontal radius times o_c; geometry uses a spheroid and a capped trunk. Crown base RGB comes from the world's vegetation reflectance at every detail level, independently of the terrain pixel under the tree. Rock-colored habitat therefore does not recolor foliage; lighting, crown texture, and atmospheric haze still modulate its displayed appearance. Supplied nonzero morphology parameters override these defaults. Missing or nonfinite environmental scaling falls back to one. This edition replaces the old ln(p/p⊕)+1 factor, which crossed zero at p=p⊕/e and became negative below it. Low gravity is also bounded to avoid divergent trunks.

The inverse gravity height trend is a visual structural-support analogy. The pressure trend favors wide flat crowns at low pressure and narrower taller crowns at high pressure. Neither is a biomechanics or evolutionary prediction; wood strength, wind, tissue density, fluid transport, and competition are not solved.

| Foliage parameter | Generated range |
|---|---|
| Cluster scale | .25–1.4 m |
| Leaf scale | .025–.18 m |
| Density | .35–.95 |
| Contrast | .30–.75 |
| Roughness | .08–.40 |
| Vertical stretch | .45–2.8 |
| Pattern | Four equally indexed pattern families |

The foliage volume is periodic in three dimensions. Trilinear interpolation uses cubic smoothstep weights and their analytic gradients; density and gradient supply matching opacity and surface shading. Fine structure is reduced with viewing footprint. Cluster and leaf lengths are scenario parameters, not a calculated response to stellar spectrum or atmospheric chemistry.

$$N(\mathbf{x}) = \sum_{c \in \{0,1\}^3} v_c \prod_{k=1}^{3} w_{c_k}(x_k), \quad w_1(t) = 3t^2 - 2t^3, \quad w_0 = 1 - w_1$$

Scenery placement is stable over geographic cells and changes with terrain material and habitat. Canopy-radius rejection removes overlapping large crowns. Distance clipping and projected detail affect which objects are drawn; they do not change the underlying life suitability calculation.

# 23. Hydrostatic scale height and cloud condensation

Implementation basis: cloudclimate.h; planet.cpp.

$$\bar{\mu} = \frac{\sum_i x_i \mu_i}{\sum_i x_i}, \quad R_s = R_g/\bar{\mu}, \quad H = R_s T/g, \quad m_{col} = p/g$$

Gas fractions are locally normalized without discarding supplied composition. Molar heat capacities approximate ideal gases, e.g. $H_2$ 28.8, He/Ar 20.8, $N_2$ 29.1, $O_2$ 29.4, $CO_2$ 37.1, $H_2O$ 33.6, $CH_4$ 35.7, and $NH_3$ 35.1 J $mol^{-1}$ $K^{-1}$. Missing mixture uses air μ=.02897 kg $mol^{-1}$ and molar heat capacity 29.1.

$$p_j = p_s e^{-0.12j}, \quad T_j = \max(T_{skin}, T_{j-1} e^{-0.12\kappa}) \quad (j = 1, \ldots, 64)$$

$$\kappa = \frac{R_g}{c_{p,mol}} \frac{1 + Lq/(R_s T)}{1 + L^2 q/(c_{p,s} R_v T^2)} \quad \text{(saturated water)}$$

Unsaturated levels use κ=R_g/c_p,mol. The moist correction uses L=2.5×$10^6$ J $kg^{-1}$ and R_v=461.5 J $kg^{-1}$ $K^{-1}$ when water is saturated between 235 and 647 K. Skin temperature is min(T_surface,max(30,.72T_eq scaled to the local temperature bin)). Seventeen surface-temperature bins span .55–1.25 of the mean. This is a vertical condensation closure, not a radiative-convective weather solution.

$$b_i = \max_j \left[ e^{-0.12j} \max\left(0, 1 - \frac{p_{sat,i}(T_j)}{x_i p_j}\right)\right]$$

The condensate availability b_i controls a column path for water/ice, $CO_2$, methane, and ammonia. For absent humidity with an ocean but no specified mixture, assumed relative humidity is .75√f_o for warm water or .20√f_o for ice; inferred vapor mole fraction is capped at .2. An explicitly dry specified mixture does not receive that fallback.

# 24. Cloud optical depth, cover, and condensate colors

Implementation basis: cloudclimate.h; cloudlighting.h; celestial.cpp.

$$\tau_c = \frac{3W_c}{2\rho_c r_{eff}}, \qquad W_i = \frac{p}{g}\frac{x_i\mu_i}{\bar{\mu}} b_i \varepsilon_i \zeta$$

W_c is condensate path in kg m⁻²; the extinction formula assumes efficiency near two for particles large compared with wavelength. Settling factor ζ=clip(√(9.80665/g),.2,3)×p/(p+1000). Particle sizes, precipitation efficiencies εᵢ, and coverage are assumed rather than observed microphysics.

| Condensate | Radius / density | Efficiency and coverage |
|---|---|---|
| Water liquid | 10+5f_o µm / 1000 kg m⁻³ | ε=.0015; coverage clipped from .12+.72√RH+.08f_o |
| Water ice | 20 µm / 917 kg m⁻³ | ε=.02; blends with water across 273.15–235 K |
| $CO_2$ | 30 µm / 1600 kg m⁻³ | ε=10⁻⁵; coverage .18 |
| Methane | 20 µm / 500 kg m⁻³ | ε=.0005; coverage .12 |
| Ammonia | 15 µm / 700 kg m⁻³ | ε=.0005; coverage .25 |
| Sulfuric acid | 1 µm / 1800 kg m⁻³ | precursor conversion 10⁻⁴; coverage .98 |

Methane saturation uses 111.66 K at one atmosphere, latent heats 9140/8200 J mol⁻¹, and critical temperature 190.56 K. Ammonia uses 239.82 K, one atmosphere, 29000/23350 J mol⁻¹, and 405.4 K critical temperature. The assumed phase switches are 90.69 and 195.4 K respectively. Optical depths sum and are capped at 100; coverage uses the largest contributing layer. The species with largest opacity selects the condensate label.

$$c_v = C((T-240)/50;0,1)\sqrt{C(RH;0,1)}\frac{p}{p+3000}$$

A warm acid layer is allowed if the vertical profile crosses 200–400 K, the surface exceeds 300 K, and $SO_2$, $H_2O$, and either $CO_2$ or $O_2$ are present. Available precursor is min(x_$SO_2$,x_$H_2O$) times .098079/µ. This is a sulfate proxy, not photochemical kinetics.

Neutral water/ice and non-acid decks use bright RGB (244,246,249) and deep (195,199,205). Acid decks draw warm bright RGB ranges (242–249,234–243,214–228) and reduce the deep channels by (.78–.86,.66–.75,.44–.56). These are display colors rather than measured complex refractive indices. Star color, illumination, shadows, and viewing depth further alter them.

# 25. Rocky-world cloud shapes and storms

Implementation basis: celestial.cpp: cloud-map generation; clouds_gpu.cpp.

$$\mathcal{R} = C\left[\frac{\omega R}{1.7\sqrt{R_s T}}; 0, 3\right], \quad A = c_v \min(1, \tau_c/3) C(\mathcal{R}; 0, 1)$$

The number of storm systems is round(U_integer(7,10)×A). One in four candidate systems is compact; the others have broad frontal heads. Angular radii are U(.08,.13) or U(.14,.24) radians, multiplied by clip(1/√(□+.15),.6,2.5). Compact latitudes have absolute range .20–.52 rad, broad systems .44–1.09 rad. Hemisphere and spin sign set winding direction. Up to 48 placement trials avoid overlapping cores.

$$r = \frac{\arccos(\mathbf{n} \cdot \mathbf{c})}{r_{storm}}, \quad \theta_{twist} = s(5 + 3U)(1 - r)^2$$

A tangent-plane east/north basis rotates by θ_twist. A smooth envelope vanishes over normalized radius .72–1. Bent fronts use distance u−b(v²−.15), b=U(.35,.80). Compact systems have a Gaussian cloud shield and an eye clearing between .035 and .11 radius; broad systems combine a head with a curved frontal tail. These are procedural weather forms; no Navier–Stokes pressure/wind evolution is solved.

$$F_b = N_5(4\mathbf{w}), \quad F_d = N_4(18\mathbf{w}), \quad F_f = N_4(9w_x, 9w_y, 36w_z)$$

Coordinates w are the storm-advected unit sphere plus .35 times low-frequency warping. Cloud bodies use smooth thresholds .48–.57 with .22(detail−.5); wisps combine .42–.50 and .49–.60 thresholds. Optical columns vary independently at scales 2.6 and 30. Cloud coverage shifts thresholds by .18×(coverage−.65); the optical scale is clip(τ_c/6,0,4). Resolution limits the finest octave to about two pixels.

$$\alpha_{body} = b(1 - e^{-d_b \tau_*}), \quad \alpha_{wisp} = 0.65 f(1 - e^{-d_f \tau_*})$$

Body depth is (.55+3.80column)(.55+.80cells); wisp depth is (.12+.65column)(.80+.40cells); storm depth is (1.20+3.20column+1.50shape)(.70+.60cells). The final opacity is the maximum of storm, background, and overcast-deck terms. A globally opaque state requires every map pixel to reach 179/255 opacity. The weather random state combines object identity, simulation hour, and weather revision; changing epoch can create another realization, not a continuous forecast.

# 26. Gaseous-world bands, vortices, and overcast decks

Implementation basis: giantclouds.h; celestial.cpp.

$$\psi = z\nu + \phi + 1.4N_b + 1.2N_f, \quad B = S\left(\frac{0.5 - 0.5\cos\psi - 0.18}{0.62}\right)$$

Spherical noise and latitude-dependent shear deform band boundaries. Broad fields use scales 3.2 and 4.7; tracers begin at scale 12, with a .12 sin(zν+φ) longitudinal shear. Three-octave folds and threads supply detail. Fine structure fades with angular pixel footprint and local vortex strain, limiting aliasing.

$$r^2 = \left[\frac{(\lambda - \lambda_v)\cos\varphi_v}{w_v}\right]^2 + \left[\frac{\varphi - \varphi_v}{h_v}\right]^2$$

$$\theta = t_v(1 - r^2/2.4^2)^3(1 + 0.23N) \quad (r < 2.4)$$

The rotation advects the bands and tracers themselves. A major storm has height U(.075,.12) rad, aspect ratio U(2.7,3.8), and twist magnitude U(3.5,5). Smaller vortices have latitude U(−1.14,1.14) rad, height U(.014,.047), aspect U(1.4,3.1), twist U(1.7,3.5), and a 40% bright variant. There are 38 small vortices in banded patterns and seven in hazy patterns. The parent map enables a major storm with a 20% draw in eligible non-synchronous worlds.

A palette is interpolated by a band coordinate displaced by broad noise, folds, and threads. Fine fields vary brightness while retaining palette hue. Polar blending suppresses contrast. Clear style returns the base color. Synchronous style shears longitude by jet strength×cos¹⁵(latitude), then transforms to a star-axis frame so band geometry bends around the illumination axis. These are static albedo patterns, not calculated gas circulation.

$$t = C[|z|^{k_p} + c(N - 0.5) + mN_m; 0, 1], \quad C_{deck} = (1 - t)C_{deep} + tC_{bright}$$

A separate overcast closure uses zonal stretching U(5,9), scale U(1.6,2.6), shear U(1.5,4), exponent U(1,1.6), warp U(.5,1.1), polar exponent U(1.8,3.6), and contrast U(.38,.62). Rotation modifies stretching; optical thickness lowers contrast. Synchronous decks use the star-axis cosine instead of poleward latitude. No inference of actual jet speeds is implied.

# 27. Giant reflectance spectra and band colors

Implementation basis: giantreflectance.cpp; giant_reflectance.json.

A 16-model reflection grid uses two atmosphere families, two enrichment values per family, and four distances (.8,2,5,10 AU), sampled at 350–1000 nm in 5 nm steps. Distance and enrichment interpolate logarithmically within a family; family mixing depends on mass. Outside the grid, endpoint clamping is reported. This library follows Cahoy, Marley, and Fortney [R10].

$$w_N = C\left[\frac{\ln(318/m)}{\ln(318/17)}; 0, 1\right]$$

$$Z = C[3(318/m)^{0.6}10^{0.25[Fe/H] + 0.3(U - 0.5)}; 1, 30]$$

$$a_{band} = a_{eq}(g/15)^{0.04}10^{0.12(U_1 - 0.5) + 0.16(U_2 - 0.5)}$$

Atmospheric enrichment Z is a scenario, not a measured bulk metallicity. Eight band realizations add cloud fractions U(.04,.16) and blue-absorber optical coefficients h×U(.3,1.7), with h=U(.08,.38) unless supplied. The equivalent Solar distance is 1/√F, or (255/T_proxy)² when a temperature proxy supplies it.

$$R_{band}(\lambda) = [(1 - c)R_g(\lambda) + 0.65c]\exp[-h_b(450/\lambda)^4]$$

For λ≥600 nm a local maximum within ±60 nm estimates a continuum, and trough transmission is raised to a methane scale. Supplied methane uses clip(x_CH₄/(.0005Z),0,4); otherwise carbon-relative-to-iron provides a bounded power-of-ten scale. This is an absorption-depth adjustment to a library, not a new chemical equilibrium calculation.

$$X_k \propto \sum_{\lambda} w_\lambda F_\odot(\lambda)\frac{B_\lambda(T_*)}{B_\lambda(5778)}R(\lambda)\bar{x}_k(\lambda)$$

Trapezoidal quadrature and CIE matching functions produce XYZ; a standard linear RGB matrix follows, with peak normalization and gamut clipping. B,V,R filter integrals relative to the reference illuminant produce photometric colors. The library is an opposition reflection model; display RGB normalization preserves hue but discards absolute reflectivity information. Use the photometric model rather than normalized RGB for brightness.

# 28. Terrain noise, rocks, craters, and molten surfaces

Implementation basis: misc.cpp; noise.cpp; celestial.cpp; rockmeshcache.h; lavafissures.h.

$$N_o(\mathbf{x}) = \frac{1}{2}\left[1 + \frac{\sum_{j=0}^{o-1} g^j n(\ell^j \mathbf{x})}{\sum_{j=0}^{o-1} g^j}\right]$$

$$Q_o(\mathbf{x}) = \frac{\sum_j g^j [1 - |n(\ell^j \mathbf{x})|]^2}{\sum_j g^j}$$

Gradient noise n uses a shuffled 256-entry permutation, lattice gradients, and quintic fade $6t^5-15t^4+10t^3$. N_o is normalized multiscale noise; Q_o folds it into ridges. Frequencies, gains, offsets, and warps describe appearance rather than measured geological power spectra. Sampling on a unit sphere makes longitude continuous.

Rocky terrain chooses five through eight octaves, feature scale U(1.5,2.9) with water or U(.2,.8) without it, gain .5 or 2.5, and frequency ratio U(.51,.53)√max(1,ln(R expressed in metres)). Domain warping and a regional roughness mask vary local gain. The metre reference in that logarithm is an implicit calibration scale, not a physical law. Color provinces, grain, mottled boundaries, and red/blue hue variation are separate spherical fields. Cold organic stains are restricted by thermal regime; they are not mineral identifications.

$$N_{crater} = \left\lfloor N_0 \frac{B}{1 + 3p/p_\oplus} \right\rfloor, \quad D = 50 + [\min(0.3R, 900000) - 50]U^5$$

$N_0$ is 300 ordinarily and three on molten surfaces. Craters are uniform in area. Depth is visual relief times U(.3,.9), rim height .15–.35 of depth, and rim width .15–.30 of angular radius. Large basins may have central peaks; rays are rare and use angular lobes. This synthetic size distribution is skewed to small craters but is not a calibrated crater chronology.

$$b_{slab} = C[\log_{10}(p/p_\oplus)/3; 0, 1], \quad a_{rock} = C[a_m(0.45 + 1.35U)(1 - 0.68 b_{slab}); 0.045, 2.2]$$

Boulders perturb a 26-vertex dome. Horizontal scales are U(.70,1.30) and U(.65,1.30); vertical scale U(.85,1.15), with small tilt and an offset apex. Three profile families produce broad caps, unequal shoulders, or taller crests. Pressure-driven slab bias is an artistic prior, not an erosion law.

Molten textures combine warped six-octave terrain, fissures, sin/cos vent masks, and lowland basins. Fissures depress relief by .08 of the normalized height; vents activate above .64 in their trigonometric field. Piecewise red/orange/gold RGB ramps encode lava activity, multiplied by a thermal emission factor. Reflected basalt remains in daylight. This display palette is not a calibrated spectroradiometer or a viscosity/convection model.

# 29. Sky colors, extinction, yellowing, and dense haze

Implementation basis: visuals.cpp; color.cpp; pressurehaze.h.

$$\mathbf{C}_{sky} \propto (1 - f_d)(0.37, 0.58, 0.81) + f_d \mathbf{C}_{planet}$$

The baseline Rayleigh-like blue vector blends with a surface-colored particulate component, modulated by daylight, starlight, and an optional city-light term. Grazing-path colors use channel exponentials with a pressure fourth-root depth clipped between .2 and 3. Apparent limb strength grows as min(1,(p/p⊕)^(1/4)). These are compressed display closures rather than a full angular multiple-scattering solution. A molten-world option supplies a warm mineral-vapor palette.

$$X(z) = \frac{1}{\cos z + 0.025}, \qquad \Delta(B - V) = 0.2(0.14)\frac{p}{101325} X(z)$$

The yellowing function mirrors zenith angles beyond 90°. The .025 denominator regularizes the horizon; the .14 mag per airmass terrestrial color excess and .2 display multiplier are empirical. Some direct-light paths apply a further factor three. Composition-specific extinction and true curved-atmosphere airmass are not evaluated here. This formula can overstate extreme-pressure color shifts; it should not be interpreted as a measured color excess.

$$\beta_{550} = \frac{p}{k_B T}(4.6 \times 10^{-31}) r_{scat} S\left(\frac{p - 2 \times 10^5}{8 \times 10^5}\right)$$

$$\beta_c = \beta_{550}[(550/\lambda_c)^4 + f_d], \quad \lambda_c = (650, 550, 450)\ \text{nm}$$

Dense pressure haze activates above $2 \times 10^5$ Pa and reaches its full smooth ramp at $10^6$ Pa. Composition-weighted approximate Rayleigh strengths set r_scat. Number density follows the ideal-gas law; dense-fluid correlations and real-gas refractivity are omitted.

$$\alpha_{sky} = 1 - e^{-\beta_{550} H}, \quad C_c(d) = C_c(0) e^{-\beta_c d} + L_c(1 - e^{-\beta_c d})$$

$$T_{diff,\,c} = (1 + 0.75 \beta_c H)^{-1}, \quad T_{display,\,c} = T_{diff,\,c}^{1/2.2}$$

Diffuse two-stream transmission approximates redistributed light before haze airlight is normalized. A luminance cap uses .2126R+.7152G+.0722B and limits haze luminance to 140 in display space. This appearance haze is separate from the thermal-opacity closure and does not itself close an anti-greenhouse energy budget.

# 30. Atmospheric refraction and the local horizon

Implementation basis: planet.cpp; surfacerefraction.h.

$$P_r = p/101325, \qquad P_r \mapsto 5 + \log_{10}(P_r - 4) \quad (P_r > 5)$$

$$k = 0.5\max(1, P_r)^{1.3}, \quad h_c = -1 + k\ln[1 + e^{(h+1)/k}]$$

$$R_{deg} = \frac{1.02}{60}\cot\left[h_c + \frac{10.3}{h_c + 5.11}\right]\frac{283}{T - 0.15}P_r$$

Angles inside the cotangent are degrees. This regularized Saemundsson-style estimate adds a shift to true altitude. It becomes zero when its angular argument reaches 90° and is clipped to 0–5°. Pressure compression and broadening prevent excessive horizon sticking. The terrestrial empirical formula is not a realistic duct/mirage solver in an arbitrary atmosphere.

$$\nu = (n-1)_{std}\frac{p}{101325}\frac{273.15}{T}, \quad \kappa_{ray} = \frac{\nu}{(1+\nu)T}\left(\frac{\mu g}{R_g} - \Gamma\right)$$

$$c = \frac{1}{2}\max(0.01/R, 1/R - \kappa_{ray}), \quad d_H = \sqrt{h_{eye}/c}, \quad \delta = \tan^{-1}(2\sqrt{h_{eye}c})$$

The near-ground model uses density-scaled refractivity and a boundary-layer lapse rate. A supplied lapse value is retained. Otherwise .0065g/9.81 K m⁻¹ blends toward .7 K m⁻¹ for dense hot atmospheres, using smooth transitions in refractivity (.0005–.015) and temperature (320–700 K). This is a dense-hot scenario inspired by local lower-atmosphere behavior, not a universal lapse law. The curvature bound limits an effective-radius singularity to 100 geometric radii. Ground geometry and distant sky refraction are separate approximations.

# 31. Photometry, surface light, cloud light, and visibility

Implementation basis: planet.cpp; color.cpp; cloudlighting.h; visuals.cpp; sphere_impostor.cpp.

$$m = M + 5\log_{10}(d/10\text{pc}), \qquad F_1/F_2 = 10^{-0.4(m_1 - m_2)}$$

$$\Phi_L(\alpha) = \frac{\sin\alpha + (\pi - \alpha)\cos\alpha}{\pi}$$

$$\Phi_{HG} = (1 - G)e^{-3.33\tan^{0.63}(\alpha/2)} + Ge^{-1.87\tan^{1.22}(\alpha/2)}$$

Planet brightness uses appropriate phase functions, albedo, size, source brightness, and source/observer distance. Solid-surface slope estimates and an opposition-surge term adjust the phase falloff; clouds can blend a Lambert-like contribution. These analytic functions approximate integrated brightness rather than solving a bidirectional reflectance distribution for every terrain element.

$$F_{refl} \propto A_g R^2 \Phi(\alpha)/(r_*^2 d_{obs}^2)$$

Intrinsic surface colors are distinct from stellar illumination. Blackbody/color-index transformations approximate the host spectrum; pressure yellowing, ambient light, cloud shadows, and haze change the result. Scenery uses diffuse plus positive surface-normal illumination; dense haze transfers part of direct light into diffuse ambient. Cloud shading uses optical thickness, sun angle, and forward/diffuse scattering approximations. Texture opacity is not a geometric cloud height measurement.

Display brightness, gamma, red-light mode, glare, bloom, angular pixel size, and exposure can change apparent hue and visibility. Therefore a screen RGB sample cannot be inverted uniquely to atmospheric composition or surface reflectance. Fine noise and distant meshes are filtered or simplified for the projected footprint; this is a sampling operation rather than a physical change in the object.

# 32. Eclipse corona filaments

Implementation basis: visuals.cpp: corona rendering.

Near-total obscuration reveals a synthetic corona around the stellar disk. The amplitude is A=clip((obscuration−.97)/.03,0,1)^1.5 and values below .02 are omitted. A bright inner halo spans 1.01–2.2 stellar radii with opacity 150A and power falloff 2.6; a faint outer halo spans 1.01–5 radii with opacity 28A and power falloff 1.7. These are display exposure choices, not an absolute coronal density inversion.

$$r(t) = 0.985R + Lt, \quad L = R(0.65 + 3.8u_1^2)[0.8 + 0.2\cos(2\theta_0 - 0.7)]$$

$$\theta(t) = \theta_0 + b(1 - e^{-3t}) + s(1 - 0.65t) + 0.012t\sin(7t + 6u_3)$$

$$w(t) = w_0[0.7 + 0.6\sin(\pi t)](1 - 0.7t), \quad a(t) = a_0 e^{-1.2t}(1 - t)^{1.3}$$

Ninety-six angular bundles contain three neighboring strands each. The normalized radial coordinate t spans zero to one with 24 intervals. Stable index hashes supply $u_1$ through $u_4$; $\theta_0$=(k+.8($u_3$−.5))2π/96, bend b=.70($u_4$−.5), and strand offset s=(j−1)(.009+.012$u_2$). Base halfwidth $w_0$=R(.008+.013$u_2$) times 1.35 for the middle strand or .65 for the others. Base alpha $a_0$=A(24+38$u_2$) times .8 for the middle strand or .55 otherwise.

Each cross-section interpolates from zero opacity at both edges to a(t) at its center. Smooth radial fading removes a sharp triangular tip; curvature and neighboring offsets produce hairlike bundles. The random pattern is fixed independently of simulation time and camera orientation. This is a plausible optical illustration of filamentary structure, not a magnetohydrodynamic reconstruction or a prediction of an individual eclipse. A thin red chromospheric rim and small prominence markers remain separate overlays.

# 33. Generated regions and star systems through Ctrl+P

Implementation basis: fictionalstars_ui.cpp; fictionalstars.cpp; object_name_pool.dat; galaxygeometry.h.

Ctrl+P opens Populate Region. Region geometry, host galaxy, stellar population choices, spacing, and a reproducible random-state value define a synthetic catalog. Generated catalog numbers are unique within the chosen prefix. Independent random streams for positions, stellar properties, multiples, planes, names, and planets reduce accidental coupling when one subsystem changes.

$$\cos\theta = 2U - 1, \quad \phi = 2\pi U, \quad r = R_{max}U^{1/3}$$

Uniform directions and the cubic-root radial law sample volume, followed by spatial-density and spacing rejection. Galaxy images can weight local populations; an analytic fallback uses exp(−3R/R_g) times .85 exp(−|z|/h)+.15 exp(−|z|/(3h)), or exp(−5r/R_g) for a spheroid. Clusters are additional spatial scenarios. These are population visualizations, not a complete selection-corrected Galactic survey.

$$v_c = v_0\frac{R}{\sqrt{R^2+R_0^2}}, \quad v_0 = C[220\sqrt{R_g/50000}\,;25,320]\,\mathrm{km\,s^{-1}}$$

$$\sigma_R = 0.16v_0[(t+0.1)/4.7]^{0.35}, \quad \sigma_z = 0.55\sigma_R, \quad \sigma_\phi = \sigma_R/\sqrt{2}$$

R_g is in light-years and $R_0$=max(100,.03R_g). A spheroid instead uses σ_R=.55$v_0$ and isotropic dispersion. Gaussian components use Box–Muller sampling. Disk streaming includes an asymmetric-drift proxy; spheroids stream at .1v_c. These kinematics are illustrative and are not propagated in a self-consistent Galactic potential.

Stellar age and spectral phase draws depend on population and blue light weighting. A lifetime proxy $10m^{-2.5}$ Gyr limits short-lived main-sequence stars. Typical main-sequence properties come from the spectral tables. Evolved phases use temperature/radius distributions and luminosity from radius and temperature; the generator does not run stellar evolution for every object. Rotation follows clip(25√(t/4.6)(T⊙/T)²,.25,150) days. Companions draw mass ratios and orbital scales, usually near a shared plane, with eccentricity below .5.

## Origin of the names

The optional proper-name pool is the repository’s UTF-8 object_name_pool.dat: curated mythological, historical, literary, and ancient-place names. Blank/comment lines and duplicates are excluded. The pool is deterministically shuffled; reserved names are avoided, and numerical suffixes permit reuse after exhaustion. Component letters and planet letters extend the host designation. These are generated labels, not IAU-approved discoveries or identifiers from a measured star catalog.

# 34. Generated planets, moons, rings, and fallback properties

Implementation basis: fictionalstars.cpp; planet.cpp; moon.cpp; orbit_spacing.h.

$$M = M_{min}(M_{max}/M_{min})^{U}, \quad M_{min} = 0.04M_{\oplus}, \quad M_{max} = \min(3M_J, 0.0003M_*)$$

The regional generator requests three through ten planets around selected hosts. Initial radius is m^.28 U(.9,1.1) Earth radii below two Earth masses, m^.35 U(1,1.9) below twenty, and min(12,3.5(m/20)^.35) U(.8,1.2) above that. These are broad priors; later classification and atmosphere scenarios can revise inferred properties.

$$a_{in} = \max(0.02\text{AU}, 8R_*), \quad h = 8\left[\frac{M_i + M_{i-1}}{3M_*}\right]^{1/3}$$

$$a_i = a_{i-1}\frac{1 + e_{i-1} + h}{1 - e_i - h}U(1.05, 1.30)$$

Eccentricities are U(0,.02), inclination U(0,2°) about the selected plane, and orbital angles uniform. The first orbit is a_in U(1,2). Outer radius is at most 500 AU and is further reduced to .05 of companion pericenter times a mass-ratio cube root. These conservative spacing filters reduce obvious overlaps; no many-body integration certifies stability.

$$M_s = U^4 M_p/4000, \quad R_s = R_{\oplus}(M_s/M_{\oplus})^{1/3}U(0.9, 1.5)$$

Moon albedo is U(.1,.6), color index U(.8,1.8), eccentricity U(0,.1)¹⁰, and starting orbit is the safe inner bound times (1.1+20U⁴). Subsequent period ratios are U(1.8,2.2). Inclinations scale as U⁴×.2×π/2 and obliquities U¹⁰×π/2, multiplied by a proximity-to-disruption alignment factor; a 3% draw makes inclination retrograde. Clearance and outer stability filters apply before acceptance.

Ring occurrence begins at .65 for massive giants, .15 for intermediate planets, and .02 for terrestrial masses; add .25 below 170 K, reject above 600 K, and subtract .60 above 300 K, then clip to [.01,.95]. Slow/resonant rotation suppresses generation. Ring material density is 1000 kg m$^{-3}$ below 200 K and 2500 otherwise. Inner radius is uniform from 1.1R to .6 Roche radius; outer radius from 1.15 times that inner radius to .98 Roche radius. Opacity is U(.1,.8)+.2 times normalized ring width, capped at one. These are appearance scenarios, not measured ring occurrence rates. Ring textures are radial profiles with distinct inner and outer endpoints. Sampling clamps those endpoints so an opaque outer edge cannot wrap into a transparent inner gap; the same rule applies to shadows on the planet. The reference mesh converts its normalized radial coordinate to the map longitude interval from −π to just below π.

General missing-radius fallbacks differ from regional initial draws: solid worlds use 1.02m^.27 Earth radii, intermediate volatile-rich worlds use max(2,.56m^.67), hot giants use an assumed-density sphere, and the remaining giant fallback is 18.6 Earth radii. That last value is a coarse visual fallback rather than a realistic universal giant mass–radius relation; explicit radii and supported envelope scenarios are preferable.

# 35. Procedural atmosphere mixtures and random reproducibility

Implementation basis: celestial.cpp; planet.cpp; noise.cpp.

Conserved inferred volatile pools and supplied mixtures outrank illustrative mixture generation. Where procedural mixtures are allowed, independent uniform draws fill trace gases and the principal gas takes the remainder; normalization enforces nonnegative fractions summing to one. Those trace ranges are priors, not equilibrium chemistry or detections.

| Scenario | Principal draws before normalization |
|---|---|
| Gas giant | He .01–.20; $CH_4$ .002–.005; $NH_3$ .0001–.0003; $C_2H_6$ 5–8 ppm; $H_2O$ 0–5 ppm; remaining $H_2$ |
| Ice giant | He .10–.30; $CH_4$ .001–.003; $NH_3$ .0001–.0003; $C_2H_6$ 5–8 ppm; $H_2O$ 0–5 ppm; $H_2$ .70–.85 |
| $CO_2$-rich | $N_2$ .01–.10; $SO_2$ .0001–.001; Ar and $H_2O$ 10–100 ppm; $H_2S$ 1–10 ppm; CO 10–30 ppm; He 10–20 ppm; remaining $CO_2$ |
| Cold $N_2$-rich | $CH_4$ .01–.10; $H_2$ .001–.005; $C_2H_6$ .00001–.01; remaining $N_2$ |

The optional terrestrial biosphere mixture draws $CH_4$ 0–.0005, HCN 0–.001, $NH_3$ 0–10 ppm, Ar 10 ppm–.005, $CO_2$ 10 ppm–.01, $H_2O$ .001–.015, with CO tied to $CO_2$ and ethane tied to methane. A 40% volcanic draw introduces $SO_2$/$H_2S$; a 3% oxygen draw introduces $O_2$ and proportional $O_3$. These probabilities are artistic assumptions, especially the oxygen frequency. They do not calculate biological oxygenation.

$$Z = \sqrt{-2\ln U_1}\cos(2\pi U_2), \qquad X_{log} = \exp[\ln l + U\ln(h/l)]$$

Uniform, Gaussian, log-uniform, discrete, and power-transformed draws recur throughout generation. Integer hash mixing creates stable local variation from object or geographic identities. Mersenne Twister streams make generated appearances repeatable for a fixed input state and operation order; exact cross-platform bitwise equivalence is not promised by every distribution implementation. The reproducibility record should include source digest, catalog version, random state, epoch, weather revision, and explicit overrides.

# 36. Stellar surface textures and further appearance estimates

Implementation basis: celestial.cpp: stellar-map generation; star.cpp; wdlimb.h; lavastatic.h; lavafissures.h.

$$\tau_{conv} = 12[C(T; 3000, 7000)/T_{\odot}]^{-2.5} \text{ days}, \quad Ro = P_{spin}/\tau_{conv}$$

$$A = \min(1, 0.35/\max(0.05, Ro)), \quad \Delta T_{spot} = \max[150, \min(0.541T - 1323, 0.45T)]$$

Ordinary cool-envelope stars below 7000 K can receive spots and granulation. Below .35 solar masses they are treated as fully convective. Spot-group count is floor(7A+U(.8,1.5)); each group has two through five members. Low Rossby number can favor polar groups; fully convective stars otherwise sample uniform surface area, and other stars favor ±5–35° active latitudes. Spot angular size starts at .35+7A degrees and receives member-dependent random factors. These are activity morphology scenarios, not magnetic-field reconstructions.

$$H_{photo} = \frac{k_B T}{1.3 m_u g}, \quad d_{granule} = \min(10 H_{photo}, 0.5R)$$

A granulation correction factor eight adjusts visible cell count; angular sampling limits the smallest cells. Temperature contrast begins at .035T times a bounded square-root temperature factor. Larger-scale network and facular gains depend on activity. Local temperatures modify color through Planck B/V ratios. Convective gravity darkening uses exponent .08, radiative-envelope exponent .25, with a bounded oblateness proxy. These fields illustrate surface inhomogeneity without solving radiative magnetohydrodynamics.

$$I(\mu)/I(1) = 1 - a(1 - \mu) - b(1 - \mu)^2$$

Quadratic limb darkening uses temperature/gravity-dependent coefficients; white-dwarf atmosphere colors and limb prescriptions can take precedence. Glare and coronal display overlays are visibility aids, not calibrated imaging predictions. Molten ground fissure patterns and thermal modulation likewise remain procedural unless an explicit measured map is supplied.

# 37. Periodic volatile exchange and cloud illumination

Implementation basis: shore.cpp; cloudlighting.h.

Seasonal atmospheric columns use 256 equal-time orbital phases. A first-order thermal response in normalized emitted flux is solved over a periodic cycle. The exact exponential step for each piecewise-constant forcing interval is used; a periodic starting state is obtained from the affine cycle map.

$$f_{j+1} = f_j + (H_j - f_j)(1 - e^{-\Delta t/\tau}), \quad T_j = T_m f_{j+1/2}^{1/4}$$

H_j combines the fixed internal fraction and eccentric-distance stellar forcing. At each temperature the same conserved three-species pool is partitioned by saturation. Finite gas/condensate exchange then relaxes toward that target, with three days for condensation and thirty days for evaporation. These are explicit mixing times, without cloud microphysics or latent-heat feedback in this particular global exchange model.

$$q_{j+1} = q_j + (q_{eq,j} - q_j)(1 - e^{-\Delta t/\tau_{ex}}), \quad q_{end} = q_{start}$$

A bounded fixed-point search establishes the periodic gas state for each species. This conserves available mass and permits different gas abundances on warming and cooling at the same temperature. Interpolation yields current gas and condensed columns, pressure, and time fractions classified as frozen, condensed-liquid, or all accessible water vapor. Those global classifications do not replace the local ocean enthalpy calculation.

## Cloud illumination and precipitation phase

$$d = C[2.5 \times 10^{-11} F_{source}/C(\sqrt{F_*}; 0.01, 1); 0, 1], \quad b = C(A_{ambient}; 0, 1)^2$$

$$L_{day} = \min(1, d + b), \quad L_{body} = 0.71\min[1, 1.3d^{0.666} + b]$$

The source-flux argument uses the renderer’s photometric scale; the coefficient is a display adaptation calibration. It is not a bolometric conversion in SI. Normalizing by the square root of instellation preserves visible cloud contrast across orbital environments. Background starlight remains separate from direct sunlight.

$$f_{filter} = C[\log_{10}(\max(1, p/p_\oplus))/2; 0, 1]$$

Cloud tint blends normalized transmitted light with normalized sky color according to haze. Filtering grows from terrestrial pressure to 100 terrestrial pressures; opaque decks force full diffuse filtering. A sunlit cumulus lobe reaches a factor 1.25 before clipping. Snow fraction is 1−clip((T−273.15)/2,0,1), permitting mixed precipitation just above freezing. This is a phase appearance proxy, not precipitation chemistry or a vertical melting-layer calculation.

# 38. Resolved cloud lobes, wisps, and local scenery

Implementation basis: clouds_gpu.cpp; surfacescatter.h; surfacebiomes.h; visuals.cpp; lavastatic.h.

The near-surface cloud view reconstructs a resolved volume from the opacity map rather than treating every cloud as a flat shell. Map coverage is the weighted RGB value .29R+.53G+.18B. Opaque-deck blending rises between .70 and .90. Angular footprint blends detailed lobes toward the unresolved bank between .025 and .16 in the renderer’s cloud coordinates.

$$s = 0.38 + 0.72m, \quad \theta_j = 2.39996323j + 6m$$

m is a local cloud-mass proxy, not kilograms. Six parent ellipsoids comprise a cluster. The central parent lies at horizontal radius zero with height −.08s; the others lie at .43s with height (.16+.13 sin(1.7θ))s. Central horizontal/vertical axes are .65s/.38s; peripheral axes are (.43+.05 sin(2.3θ))s and (.38+.06 cos(1.9θ))s. The 2.39996323 rad increment distributes lobes quasi-uniformly around the cluster.

$$z_k = 1 - 2(k + 0.5)/8, \quad \mathbf{u}_k = (\sqrt{1 - z_k^2}\cos\theta_k, \sqrt{1 - z_k^2}\sin\theta_k, z_k)$$

Each parent carries eight billows at parent center plus .55 times its axes times u_k. Their relative radius is .52+.035 sin(2.7θ_k+j), with a small vertical adjustment. Four finer lobes per billow use a .40 directional offset and radius .67+.025 sin(1.7θ). Lower billows are compressed while their tops remain in place, forming staggered flat bases. The mass-dependent floor is −s times a smooth blend from .80 to .20 across m=.35–.70. Ellipsoid/ray quadratics find silhouettes and distance; noise and a pixel-footprint taper soften edges.

$$\alpha = \alpha_{solid} + (1 - \alpha_{solid})\alpha_{bank}$$

This opacity union preserves the unresolved bank behind soft lobe edges. It prevents small-scale geometry from opening artificial gaps in an opaque cloud mass. Wisps use circular longitude flow, a bent latitude crosswind coordinate, and value noise at transverse scales 18 and 43. Wispiness is .001×10^(3w), with w a clipped spherical gradient-noise value in [0,1]. Veil, strand, and fiber thresholds bound cirrus opacity; the cirrus contribution fades out as coverage rises from .55 to .80. This is geometrical cloud morphology, not fluid turbulence or droplet transport.

## Scenery sizes, boundaries, and local waves

$$L = L_{min} + (L_{max} - L_{min})U^3$$

Rock and plant base size draws favor small instances. Six cube-face grids cover the sphere; geographic identity fixes candidate positions independently of camera travel. Crown exclusion compares chord distance squared with $(r_i+r_j+gap)^2$; the lower numerical identity wins a conflict. The default gap is .1 m. These procedures make populations repeatable, not ecologically self-organized.

$$\Delta\mathbf{r} = 32\mathbf{n}(\mathbf{r}/160) + 6\mathbf{n}(\mathbf{r}/36)$$

Local biome coordinates receive independent Cartesian noise displacements at 160 and 36 m scales and amplitudes 32 and 6 m. Normalization back onto the sphere and conversion to latitude/longitude break up overly smooth material boundaries without polar seams. It is not a computed erosion or vegetation-spread process.

$$W(u, v, t) = 0.6\sin(0.6u + 0.3v - 2.4t) + 0.4\sin(-0.3u + 0.7v - 1.8t)$$

The local ocean crest display displaces projected rows by this two-wave field times a projected amplitude. It draws highlights above a crest threshold and adds higher-frequency sinusoidal sparkles. The time variable follows display animation; this is a visual wave pattern without a wind-driven spectrum, gravity-wave dispersion relation, or bathymetric shoaling calculation. It must not be used to infer sea state or wind speed.

Local glowing fissures use deterministic segmented paths: an initial radius 2.2+.18×view_range×U^1.8, heading with a bounded perturbation, and fourteen through nineteen segments. Main-path turns span ±48°, branch turns ±35°, with branch starts at segments 3, 7, and 11. Brightness pulses as .88+.12 sin(2.3t+phase). This is a display of cracking/molten material, not an integrated stress or magma-flow model.

# 39. Numerical coefficients for stellar and formation inference

Implementation basis: star.cpp; stellarevolution.h; chemistry.cpp; classification_model.json.

For the central stellar bolometric-correction polynomial let $x=\log_{10}(T/K)$, and evaluate the coefficient list from the constant term upward. The following table gives the full implemented values, rather than a new fitted approximation.

| Range | Coefficients of $x^0$, $x^1$, ... |
|---|---|
| $x < 3.70$ | −19053.7291496456; 15514.4866764412; −4212.78819301717; 381.476328422343 |
| $3.70 \le x < 3.90$ | −37051.0203809015; 38567.2629965804; −15065.1486316025; 2617.24637119416; −170.623810323864 |
| $x \ge 3.90$ | −118115.450538963; 137145.973583929; −63623.3812100225; 14741.2923562646; −1705.87278406872; 78.8731721804990 |

The lower-temperature and above-50000 K continuations described earlier replace the polynomial outside their specified regimes. Main-sequence interpolation uses a continuous spectral index through the tabulated stellar sequence; endpoint handling and spectral parsing select its valid range.

$$d^2 = \left(\frac{\Delta \log_{10} T}{.035}\right)^2 + \left(\frac{\Delta \log_{10} L}{.20}\right)^2 + \left(\frac{\Delta [Fe/H]}{.40}\right)^2$$

$$\ln w = -\tfrac{1}{2}[d^2 - d_{min}^2 + P_M + P_t] + \ln(10)(\log_{10} t_{yr} - 9) - 2.35 \ln M_{initial}$$

Evolution-grid weights combine an HR-diagram distance, optional mass and age penalties, and a constant-linear-time star-formation prior with a Salpeter initial-mass factor for ordinary stellar phases. The mass penalty is squared log-mass residual divided by .30, or .04 for the compact-remnant grouping; the age penalty uses .30 dex. Medians summarize mass and age, the .1/.9 quantiles of model age bounds define the reported range, and summed weights choose the phase. Cooling tracks do not use the ordinary population prior. These are model-weighted summaries, not a universal Bayesian posterior.

$$z_f = 2\ln(m/4.37) - 0.5\ln[1 + \max(0, F - F_c)/F_s] + 3[Fe/H] + \Delta_{chem}$$

The default cool-formation envelope log-odds use a 4.37 Earth-mass transition, log-mass coefficient two, metallicity coefficient three, and irradiation coefficient .5. F_c=1 and F_s=100 Earth flux in the current scenario settings. Chemical shift is clipped to ±1.5 and offsets iron-only abundance using refractory enrichment and a water-rich alternative. Logistic transformation produces a scenario score. At overlapping calibrated fluxes, this joins the empirical class-balanced classifier; outside the fit domain it remains an explicit formation assumption. The JSON source is included in the digest appendix because coefficients, limits, and calibration settings can change.

# 40. Worked examples and sensitivity

Implementation basis: equations in this reference; implementation inputs stated below.

## Terrestrial mass and radius

$$M = 5.9722 \times 10^{24}\ \mathrm{kg}, \quad R = 6.371 \times 10^{6}\ \mathrm{m} \quad \Rightarrow \quad g \simeq 9.82, \quad v_{esc} \simeq 11.19\ \mathrm{km\,s^{-1}}$$

$$f_{atm} = 10^{-6} \quad \Rightarrow \quad p = f_{atm}\frac{GM^2}{4\pi R^4} \simeq 1.15 \times 10^{5}\ \mathrm{Pa}$$

This illustrates the hydrostatic inventory scale; it is not the full Earth solver result. Release, escape, weathering, condensation, and supplied data alter the accessible gas fraction. At unit stellar flux and A_B=.30, equilibrium temperature is about 254.6 K. With gray τ=.35, it becomes approximately 287.7 K before other constraints. This example supplies τ rather than deriving it from an assumed composition.

## Water and relief

$$f_{W,\,surface} = 2.3 \times 10^{-4} \quad \Rightarrow \quad d_W \simeq 2693\ \mathrm{m}$$

At 2500 m RMS Gaussian relief, solve zΦ(z)+φ(z)=1.077. This yields z about .99 and coverage about .84. These figures illustrate the chosen relief prior; Earth’s observed basin geometry is different. Multiplying water inventory by ten does not multiply coverage by ten because coverage is bounded while ocean depth can grow.

## Vegetation scaling

| Pressure in Earth units | p_v | Crown radius multiplier | Vertical/horizontal ratio |
|---|---|---|---|
| 0 | 0.250 | 10.000 | 0.200 |
| 0.1 | 0.250 | 10.000 | 0.200 |
| 1 | 1.000 | 2.500 | 0.800 |
| 10 | 3.459 | 0.723 | 2.768 |
| 100 | 4.000 | 0.625 | 3.200 |

All values are positive and bounded, including at p=0 and at p⊕/e. This is a repair of geometry, not evidence that vegetation can grow in vacuum. Habitat and life gates still determine whether plants are placed.

## Scientific interpretation

At fixed albedo, T_eq scales as F^(1/4); a 10% flux increase raises it about 2.4%. Atmospheric weight at fixed mass fraction scales as M²/R⁴. Cloud optical depth at fixed condensate mass scales inversely with particle radius. These sensitivities explain why uncertain radii, inventories, aerosol sizes, and albedos can dominate small changes in catalog precision.

# 41. Audit changes and scientific limitations

Implementation basis: audit edition, 13 November 2026.

Two behavior corrections accompany this reference. First, bounded positive environmental multipliers replace vegetation crown scaling that diverged and changed sign below p⊕/e; low-gravity trunk scaling is also finite. Second, a Mercury-like 3:2 spin state now has rotation period 2P_orb/3 rather than 1.5P_orb. Neither change alters supplied nonzero morphology parameters or supplied rotation periods through these default-estimation branches.

Validation: the two added morphology/rotation regression tests and all four existing thermal-opacity tests pass. Of nineteen focused volatile/rotation/morphology checks, eighteen pass. The cold-moon pressure check returns approximately 244575 Pa for its Titan-like case against a 120000–180000 Pa target, using an inferred host age of approximately 5.85 Gyr. This case does not call the altered default rotation or morphology calculations; the volatile solver and thermal-opacity calibration were not modified. The mismatch is a recorded calibration limitation, not a successful pressure prediction.

The Titan-calibrated negative ethane term remains unchanged. It is documented as a net empirical cooling proxy, with no claim that molecular ethane has negative infrared opacity. Preserving this calibration avoids an unjustified change to a coupled approximation. A future physical replacement would separate shortwave aerosol absorption/scattering, molecular longwave opacity, and vertical temperature structure, then validate the combined budget.

Other broad simplifications are explicit: one principal stellar light center; approximate bolometric and Bond-albedo extrapolation; gray greenhouse coefficients; effective volatile escape; assumed basin relief and interior storage; constant-latent-heat saturation; no general high-pressure water phase diagram; static procedural weather; speculative biosphere probabilities; and artistic pressure-dependent rock/plant forms. None should be presented as an observational inference with a calibrated uncertainty interval.

This reference covers the scientific and procedural model families that establish inferred physical properties and appearance. It does not claim a universal physical justification for decorative constants, display sampling choices, or synthetic names. Source locators and digests below permit a scientist to identify the exact audited implementation and distinguish a later revision.

# 42. Primary sources and calibration context

Implementation basis: published articles and reference data, checked 13 November 2026.


R1. Zahnle & Catling (2017), The cosmic shoreline. Atmospheric retention trend and XUV scaling motivation.

https://arxiv.org/abs/1702.03386

R2. Lopez & Fortney (2014), Understanding the Mass–Radius Relation for Sub-Neptunes. Envelope thickness fit; published domain stated in the envelope section.

https://arxiv.org/abs/1311.0329

R3. Eisenman & Wettlaufer (2009), Nonlinear threshold behavior during the loss of Arctic sea ice. Enthalpy and conductive-ice method inspiration; application parameters here are scenarios.

https://arxiv.org/abs/0812.4777

R4. NIST Chemistry WebBook, SRD 69, ethane infrared spectrum. Establishes molecular infrared absorption, not the app’s fitted net climate response.

https://webbook.nist.gov/cgi/cbook.cgi?ID=C74840&Index=0&Type=IR-SPEC

R5. Torres (2010), On the Use of Empirical Bolometric Corrections for Stars. Corrected Flower coefficients.

https://arxiv.org/abs/1008.3913

R6. Leconte et al. (2010), Is tidal heating sufficient to explain bloated exoplanets? Full eccentricity equilibrium-tide functions.

https://doi.org/10.1051/0004-6361/201014337

R7. Santos et al. (2017), Constraining planet structure from stellar chemistry. Stoichiometric condensate motivation.

https://arxiv.org/abs/1711.00777

R8. NIST Chemistry WebBook, nitrogen phase-change data; Fray & Schmitt solid-vapor fits as reproduced by Marboeuf et al. (2014).

https://webbook.nist.gov/cgi/cbook.cgi?ID=C7727379&Mask=4

R9. Kopparapu et al. (2014), Habitable Zones Around Main-Sequence Stars: Dependence on Planetary Mass. Table 1 and polynomial limits.

https://arxiv.org/abs/1404.5292

R10. Cahoy, Marley & Fortney (2010), Exoplanet albedo spectra and colors as a function of planet phase, separation, and metallicity. Reflection library context.

https://doi.org/10.1088/0004-637X/724/1/189


Citations motivate the named physical procedures only. They do not validate every numerical default, extrapolation, artistic probability, or renderer in this application. The implementation formulas and calibration qualifications in this reference govern what the app actually does.

# 43. Source traceability and reproduction

Implementation basis: repository files at document build time.

Digest algorithm: SHA-256, shown as the first sixteen hexadecimal characters. Paths in this appendix are source locators for scientific reproducibility, not a description of software architecture. Build the PDF with Python 3 plus ReportLab, Matplotlib, and NumPy using docs/build_calculation_reference.py from the repository root. DejaVu fonts and rendered 330 dpi equations are embedded; no markup interpretation is required of the reader.

| Source locator | SHA-256 prefix |
|---|---|
| src/classes/shore.cpp | bb430861d11558ce |
| src/classes/shore.h | 7d60707afb2b5774 |
| src/classes/planet.cpp | d22b64ef3f23bd79 |
| src/classes/celestial.cpp | 3cf1753a0464b0fb |
| src/classes/star.cpp | bf63e592517c0a21 |
| src/classes/misc.cpp | 152315c059eb1ebf |
| src/classes/color.cpp | 1d06e52eaf7697d9 |
| src/classes/chemistry.cpp | d3d562122e729e0d |
| src/classes/patch.cpp | f48c22c05b1eece4 |
| src/classes/climate.h | dd010f626f3a3f31 |
| src/classes/thermalclimate.h | 23caede86b6e3997 |
| src/classes/iceclimate.h | b1418fdda76f06d7 |
| src/classes/salinity.h | bf8f77f28d10af5e |
| src/classes/surfaceclimate.h | 5d9f2eab6b1288dd |
| src/classes/vegetation.h | 92622ed67e9a8922 |
| src/classes/plantform.h | 4db1840088eb8533 |
| src/classes/plantmesh.h | 63ac3e57cf715ef5 |
| src/classes/planttexture.h | f4b9fe66f75814e7 |
| src/classes/rockmeshcache.h | d284069523a45a06 |
| src/classes/cloudclimate.h | efc5f2d9fdcff0c0 |
| src/classes/giantclouds.h | a49984f107c0463a |
| src/classes/giantreflectance.cpp | b8b35cd7f274c276 |
| src/classes/pressurehaze.h | d578ef5b2f6fbb35 |

| Source locator | SHA-256 prefix |
| --- | --- |
| src/classes/surfacerefraction.h | 62784b3c1a1b18e4 |
| src/classes/tidalheat.h | 4ec39223bf8411b7 |
| src/classes/habitablezone.h | 158a65287be4ce7b |
| src/classes/noise.cpp | 4192ce7f1c23e2e9 |
| src/classes/stellarevolution.h | 2df2954763ba8a09 |
| src/classes/cloudlighting.h | cd94a52aaa9d04c5 |
| src/classes/surfacescatter.h | 9f03ea268a2a4c7b |
| src/classes/surfacebiomes.h | 7f04e553eea35922 |
| src/classes/lavastatic.h | d61deae0e2629bf2 |
| src/classes/clouds_gpu.cpp | 776a58f308a00420 |
| src/classes/fictionalstars.cpp | 209930f91cdcd9b2 |
| src/fictionalstars_ui.cpp | 1f30bf3404cb5614 |
| src/visuals.cpp | 3db55799dc60aff3 |
| src/sphere_impostor.cpp | e3a532718c9f850f |
| src/alienorum.cpp | 66df14bd899b3f17 |
| object_name_pool.dat | 53dab0edda46bd39 |
| catalogs/NASA/classification_model.json | 031a9f167677132d |
| catalogs/giant_reflectance.json | d4ba2290f722102f |